

而 $\vec{a}$ 與 $\vec{b}$ 的方向相反, 則 $\vec{a} = -4\vec{b}$

$$\Rightarrow (x, y) = -4(-5, 12) = (20, -48)$$

$$\Rightarrow x = 20, y = -48$$

$$\text{因此 } x + y = 20 + (-48) = -28$$

範例 3

類題 1

$$\because |\vec{a}| = 5, |\vec{b}| = 13 \text{ 且 } \cos \theta = \frac{33}{65}$$

$$\therefore \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = 5 \times 13 \times \frac{33}{65} = 33$$

$$\begin{aligned} & (4\vec{a} - \vec{b}) \cdot (2\vec{a} + \vec{b}) \\ &= (4\vec{a}) \cdot (2\vec{a}) + (4\vec{a}) \cdot \vec{b} - \vec{b} \cdot (2\vec{a}) - \vec{b} \cdot \vec{b} \\ &= 8|\vec{a}|^2 + 4\vec{a} \cdot \vec{b} - 2\vec{a} \cdot \vec{b} - |\vec{b}|^2 \\ &= 8|\vec{a}|^2 + 2\vec{a} \cdot \vec{b} - |\vec{b}|^2 \\ &= 8 \times 5^2 + 2 \times 33 - 13^2 = 200 + 66 - 169 = 97 \end{aligned}$$

類題 2

$$\because \vec{a} \text{ 與 } \vec{b} \text{ 之夾角為 } 60^\circ, \text{ 且 } |\vec{a}| = |\vec{b}| = 1$$

$$\therefore \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos 60^\circ = 1 \times 1 \times \frac{1}{2} = \frac{1}{2}$$

$$\begin{aligned} \text{而 } \vec{a} \cdot (-\vec{a} + 2\vec{b}) &= \vec{a} \cdot (-\vec{a}) + \vec{a} \cdot (2\vec{b}) \\ &= -|\vec{a}|^2 + 2\vec{a} \cdot \vec{b} = -1^2 + 2 \times \frac{1}{2} = 0 \end{aligned}$$

因此 $\vec{a}$ 和 $(-\vec{a} + 2\vec{b})$ 的夾角是 90°
($\because$ 兩向量垂直, 其內積為 0)



最素傳說

P.67

範例
(B)

類題 1
(A)

類題 2

類題 1

設阿樊的左右手拉力分別是 $\vec{a}$ 、 $\vec{b}$,

其重量是 $\vec{c}$ 則 $|\vec{a}| = 36, |\vec{b}| = 48$

如圖呈現平衡狀態

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

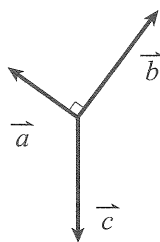
$$\Rightarrow \vec{a} + \vec{b} = -\vec{c}$$

$$\Rightarrow |\vec{a} + \vec{b}| = |\vec{c}|$$

$$\begin{aligned} \Rightarrow |\vec{c}|^2 &= |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \\ &= 36^2 + 0 + 48^2 = 3600 \end{aligned}$$

$$(\because \vec{a} \perp \vec{b} \therefore \vec{a} \cdot \vec{b} = 0)$$

$$\Rightarrow |\vec{c}| = \sqrt{3600} = 60$$



故阿樊的體重是 60 公斤

歷屆試題				P.68
1. (C)	2. (B)	3. (B)	4. (B)	
5. (A)	6. (B)	7. (D)	8. (C)	
9. (C)	10. (A)	11. (D)	12. (B)	
13. (D)	14. (B)	15. (D)	16. (C)	

解析:

$$1. \vec{AB} + 2\vec{AC} = \vec{CD}$$

$$\Rightarrow (B - A) + 2(C - A) = (D - C)$$

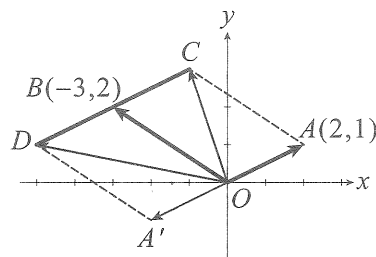
$$\Rightarrow D = -3A + B + 3C$$

$$= -3(3, 1) + (2, -3) + 3(7, -1)$$

$$= (-9, -3) + (2, -3) + (21, -3) = (14, -9)$$

故 $x = 14, y = -9$, 因此 $x + y = 14 + (-9) = 5$

2.



(1) 如圖

以 $\vec{OA}$ 與 $\vec{OB}$ 為兩鄰邊作平行四邊形 $OACB$,
則 $\vec{OB} + \vec{OA} = \vec{OC}$

(2) 如圖, 令 $\vec{OA}' = -\vec{OA}$,

以 $\vec{OA}'$ 與 $\vec{OB}$ 為兩鄰邊作平行四邊形 $OA'DB$
則 $\vec{OB} + \vec{OA}' = \vec{OD}$, 也就是 $\vec{OB} + (-\vec{OA}) = \vec{OD}$

(3) 若 $\vec{OP} = \vec{OB} + t\vec{OA}$, 其中 $-1 \leq t \leq 1$

則所有滿足 P 點所形成的線段為 $\vec{CD}$

$$\text{而 } \vec{CD} = 2\vec{BC} = 2\vec{OA}$$

$$\text{又 } \vec{OA} = \sqrt{(2-0)^2 + (1-0)^2} = \sqrt{5}$$

$$\text{故所求的線段長 } \vec{CD} = 2 \times \sqrt{5} = 2\sqrt{5}$$

$$3. \because \vec{a} \perp \vec{b} \therefore \vec{a} \cdot \vec{b} = 0$$

$$\begin{aligned} |\vec{a} - 2\vec{b}|^2 &= |\vec{a}|^2 - 2\vec{a} \cdot (2\vec{b}) + (2|\vec{b}|)^2 \\ &= |\vec{a}|^2 - 4\vec{a} \cdot \vec{b} + 4|\vec{b}|^2 \\ &= 1^2 - 4 \times 0 + 4 \times 2^2 = 17 \end{aligned}$$

$$\text{故 } |\vec{a} - 2\vec{b}| = \sqrt{17}$$

4. $\because \vec{u}$ 與 $\vec{v}$ 垂直

$$\therefore (1, 1) \cdot (x+4, y-1) = 0$$

$$\Rightarrow 1 \times (x+4) + 1 \times (y-1) = 0$$

$$\Rightarrow x + y = -3 \cdots \textcircled{1}$$

$\therefore \vec{u}$ 與 $\vec{w}$ 平行

$$\therefore \frac{1}{2x} = \frac{1}{y} \Rightarrow 2x - y = 0 \dots (2)$$

$$(1) + (2) \Rightarrow 3x = -3 \Rightarrow x = -1$$

$$x = -1 \text{ 代回 (1), 得 } -1 + y = -3 \Rightarrow y = -2$$

$$5. \therefore |\vec{a}| = 2, |\vec{b}| = 3 \text{ 且夾角為 } 60^\circ$$

$$\therefore \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos 60^\circ = 2 \times 3 \times \frac{1}{2} = 3$$

$$\begin{aligned} & (3\vec{a} - \vec{b}) \cdot (\vec{a} + 2\vec{b}) \\ &= (3\vec{a}) \cdot (\vec{a}) + (3\vec{a}) \cdot (2\vec{b}) - \vec{b} \cdot \vec{a} \\ & \quad - \vec{b} \cdot (2\vec{b}) \end{aligned}$$

$$= 3|\vec{a}|^2 + 6\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{b} - 2|\vec{b}|^2$$

$$= 3|\vec{a}|^2 + 5\vec{a} \cdot \vec{b} - 2|\vec{b}|^2$$

$$= 3 \times 2^2 + 5 \times 3 - 2 \times 3^2 = 12 + 15 - 18 = 9$$

$$6. (1) \therefore \vec{a} \cdot \vec{c} = 0 \quad \therefore \vec{a} \perp \vec{c}$$

$$(2) \therefore \vec{b} \cdot \vec{c} = 0 \quad \therefore \vec{b} \perp \vec{c}$$

$$(3) \therefore \vec{a} \cdot \vec{b} < 0$$

$\therefore \vec{a}$ 與 $\vec{b}$ 的夾角為鈍角或 180°

由(1)、(2)和(3)，可以畫出
 $\vec{a}$ 、 $\vec{b}$ 、 $\vec{c}$ 的相對位置

故 $\vec{a}$ 與 $\vec{b}$ 反向，夾角為 180°

$$\begin{aligned} \text{因此 } \vec{a} \cdot \vec{b} &= |\vec{a}| \times |\vec{b}| \times \cos 180^\circ \\ &= 5 \times 12 \times (-1) = -60 \end{aligned}$$

$$7. (1) \therefore |\vec{a} - \vec{b}|^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$$

$$= |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$$

$$\text{且 } |\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2$$

$$\therefore |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2$$

$$\Rightarrow -2\vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$(2) \vec{a} \cdot \vec{b} = (2x+1, -3) \cdot (3, x-2)$$

$$= (2x+1) \times 3 + (-3) \times (x-2)$$

$$= (6x+3) + (-3x+6) = 3x+9$$

由(1)、(2)可知，

$$3x+9=0 \Rightarrow 3x=-9 \Rightarrow x=-3$$

8. (1) 線段 $\overline{AB}$ 的中點

$$D = \frac{A+B}{2} = \frac{(x,y) + (2,0)}{2}$$

$$= \frac{(x+2,y)}{2} = \left(\frac{x+2}{2}, \frac{y}{2} \right)$$

線段 $\overline{BC}$ 的中點

$$E = \frac{B+C}{2} = \frac{(2,0) + (0,0)}{2} = \frac{(2,0)}{2} = (1,0)$$

線段 $\overline{AC}$ 的中點

$$F = \frac{A+C}{2} = \frac{(x,y) + (0,0)}{2} = \frac{(x,y)}{2} = \left(\frac{x}{2}, \frac{y}{2} \right)$$

$$(2) \text{ 向量 } \overrightarrow{DE} = E - D = (1,0) - \left(\frac{x+2}{2}, \frac{y}{2} \right)$$

$$= \left(-\frac{x}{2}, -\frac{y}{2} \right)$$

$$\text{向量 } \overrightarrow{DF} = F - D = \left(\frac{x}{2}, \frac{y}{2} \right) - \left(\frac{x+2}{2}, \frac{y}{2} \right)$$

$$= (-1, 0)$$

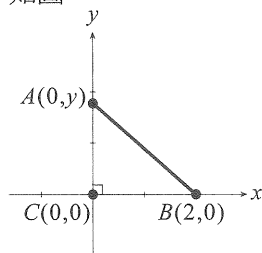
$$\text{內積 } \overrightarrow{DE} \cdot \overrightarrow{DF} = \left(-\frac{x}{2}, -\frac{y}{2} \right) \cdot (-1, 0)$$

$$= -\frac{x}{2} \times (-1) + \left(-\frac{y}{2} \right) \times 0 = \frac{x}{2}$$

$$\therefore \overrightarrow{DE} \cdot \overrightarrow{DF} = 0 \quad \therefore \frac{x}{2} = 0 \Rightarrow x = 0$$

則 A 點坐標為 $(0, y)$ 且 $y \neq 0$

(3) 如圖



點 $A(0, y)$ 位在 y 軸上 (不在原點)

因此 $\angle BCA = 90^\circ$ ，故選(C)

9. $\triangle ABC$ 之中， $\angle A = 90^\circ$

(A) 以 $\overline{AB}$ 、 $\overline{AC}$ 為鄰邊，

作一矩形 $ABDC$

$$\text{則 } \overrightarrow{AB} + \overrightarrow{AC} = \overrightarrow{AD} \neq \overrightarrow{BC}$$

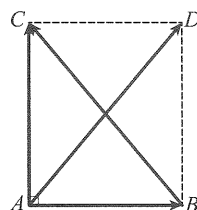
($\because \overrightarrow{AD}$ 與 $\overrightarrow{BC}$ 方向不同)

(B) $\therefore$ 三角形的任意兩邊長的和 $>$ 第三邊長

$$\therefore \overline{AB} + \overline{AC} > \overline{BC}$$

$$\text{又 } |\overrightarrow{AB}| = \overline{AB}, |\overrightarrow{AC}| = \overline{AC}, |\overrightarrow{BC}| = \overline{BC}$$

$$\text{即 } |\overrightarrow{AB}| + |\overrightarrow{AC}| > |\overrightarrow{BC}|$$



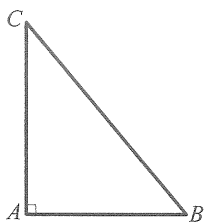
$$(C) \because \angle A = 90^\circ$$

$$\therefore \text{內積 } \overrightarrow{AB} \cdot \overrightarrow{AC} = 0$$

$$(D) \because |\overrightarrow{AB}| = \overline{AB} \neq 0$$

$$\text{且 } |\overrightarrow{AC}| = \overline{AC} \neq 0$$

$$\therefore |\overrightarrow{AB}| \cdot |\overrightarrow{AC}| \neq 0$$

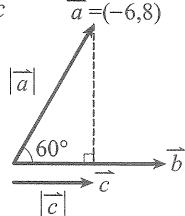


10. 如圖, $\vec{a}$ 在 $\vec{b}$ 上的正射影為 $\vec{c}$

$$\text{而 } |\vec{a}| = \sqrt{(-6)^2 + 8^2} = 10$$

則正射影長

$$|\vec{c}| = |\vec{a}| \times |\cos 60^\circ| = 10 \times \frac{1}{2} = 5$$



$$11. \begin{vmatrix} x & 1 \\ y & 2 \end{vmatrix} = 5 \Rightarrow 2x - y = 5 \dots \textcircled{1}$$

$$\begin{vmatrix} x & 2y \\ 1 & 1 \end{vmatrix} = 0 \Rightarrow x - 2y = 0 \dots \textcircled{2}$$

$$\textcircled{1} \times 2 - \textcircled{2} \Rightarrow 3x = 10 \Rightarrow x = \frac{10}{3}$$

$$\text{把 } x = \frac{10}{3} \text{ 代入 } \textcircled{1}, \text{ 得 } \frac{20}{3} - y = 5 \Rightarrow y = \frac{5}{3}$$

$$\text{因此 } x + y = \frac{10}{3} + \frac{5}{3} = 5$$

12. 由柯西不等式知

$$(x^2 + y^2)[1^2 + (-2)^2] \geq (x - 2y)^2$$

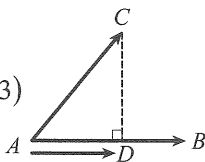
$$\Rightarrow (x^2 + y^2) \times 5 \geq 10^2 \Rightarrow x^2 + y^2 \geq 20$$

故 $x^2 + y^2$ 之最小值為 20

$$13. \overrightarrow{AC} = C - A = (5, 2) - (1, 1) = (4, 1)$$

$$\overrightarrow{AB} = B - A = (5, -2) - (1, 1) = (4, -3)$$

$\overrightarrow{AC}$ 在 $\overrightarrow{AB}$ 上的正射影



$$\overrightarrow{AD} = \left(\frac{\overrightarrow{AC} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|^2} \right) \overrightarrow{AB} = \frac{(4, 1) \cdot (4, -3)}{(\sqrt{4^2 + (-3)^2})^2} (4, -3)$$

$$= \frac{4 \times 4 + 1 \times (-3)}{5^2} (4, -3) = \frac{13}{25} (4, -3)$$

$$= \left(\frac{13}{25} \times 4, \frac{13}{25} \times (-3) \right) = \left(\frac{52}{25}, -\frac{39}{25} \right)$$

$$\therefore \overrightarrow{CD} = \overrightarrow{CA} + \overrightarrow{AD} = -\overrightarrow{AC} + \overrightarrow{AD}$$

$$= (-4, -1) + \left(\frac{52}{25}, -\frac{39}{25} \right)$$

$$= \left(-4 + \frac{52}{25}, -1 + \left(-\frac{39}{25} \right) \right) = \left(-\frac{48}{25}, -\frac{64}{25} \right)$$

$$\therefore \overrightarrow{DC} = -\overrightarrow{CD} = \left(\frac{48}{25}, \frac{64}{25} \right)$$

$$\text{則 } x = \frac{48}{25}, y = \frac{64}{25}, \text{ 因此 } x + y = \frac{48}{25} + \frac{64}{25} = \frac{112}{25}$$

14. 設 $A(x_1, y_1), B(x_2, y_2)$

$$\text{則 } 3x_1 + 4y_1 = 5, 3x_2 + 4y_2 = 5$$

$$\overrightarrow{AB} = B - A = (x_2, y_2) - (x_1, y_1) = (x_2 - x_1, y_2 - y_1)$$

$$\therefore \overrightarrow{AB} = (a, b) \therefore a = x_2 - x_1, b = y_2 - y_1$$

$$6a + 8b - 5 = 6(x_2 - x_1) + 8(y_2 - y_1) - 5$$

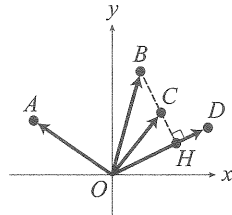
$$= 6x_2 - 6x_1 + 8y_2 - 8y_1 - 5$$

$$= (6x_2 + 8y_2) - (6x_1 + 8y_1) - 5$$

$$= 2(3x_2 + 4y_2) - 2(3x_1 + 4y_1) - 5$$

$$= 2 \times 5 - 2 \times 5 - 5 = -5$$

15. 設 B 點對 $\overrightarrow{OD}$ 的投影點是 H



(1) $\because \overrightarrow{OD}$ 與 $\overrightarrow{OA}$ 的夾角 $\angle AOD > 90^\circ$

$\therefore \overrightarrow{OD}$ 與 $\overrightarrow{OA}$ 的內積是負值, 即 $a < 0$

(2) $\overrightarrow{OD}$ 與 $\overrightarrow{OB}$ 的內積:

$$b = (\overrightarrow{OD} \text{ 長}) \times (\overrightarrow{OB} \text{ 在 } \overrightarrow{OD} \text{ 的投影量}) \\ = \overrightarrow{OD} \times \overrightarrow{OH} > 0$$

(3) $\overrightarrow{OD}$ 與 $\overrightarrow{OC}$ 的內積:

$$c = (\overrightarrow{OD} \text{ 長}) \times (\overrightarrow{OC} \text{ 在 } \overrightarrow{OD} \text{ 的投影量}) \\ = \overrightarrow{OD} \times \overrightarrow{OH} > 0$$

(4) $\overrightarrow{OD}$ 與 $\overrightarrow{OD}$ 的內積: $d = |\overrightarrow{OD}|^2 = \overrightarrow{OD} \times \overrightarrow{OD} > 0$

$$\therefore \overrightarrow{OD} > \overrightarrow{OH} \therefore \overrightarrow{OD} \times \overrightarrow{OD} > \overrightarrow{OD} \times \overrightarrow{OH}$$

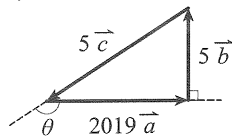
故 $d > b = c > 0$

而 $a < 0$, 因此 $d > b = c > a$

16. $\because \vec{a}$ 與 $\vec{b}$ 的夾角為 90°

$\therefore 2019\vec{a}$ 與 $5\vec{b}$ 的夾角也是 90°

以 $2019\vec{a} + 5\vec{b} + 5\vec{c} = \vec{0}$ 來作圖



$\therefore \vec{a}$ 與 $\vec{c}$ 的夾角為 θ

$\therefore 2019\vec{a}$ 與 $5\vec{c}$ 的夾角也是 θ

如圖, 故 $90^\circ < \theta < 180^\circ$